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01

a project of one's own

june 2021
paulgraham.com/own.html

so where exactly is the edge of projects of one's own? that's an interesting question, partly because the answer is so complicated, and partly because there's so much at stake. there turn out to be two senses in which work can be one's own: 1) that you're doing it voluntarily, rather than merely because someone told you to, and 2) that you're doing it by yourself.

the edge of the former is quite sharp. people who care a lot about their work are usually very sensitive to the difference between pulling, and being pushed, and work tends to fall into one category or the other. but the test isn't simply whether you're told to do something. you can choose to do something you're told to do. indeed, you can own it far more thoroughly than the person who told you to do it.

for example, math homework is for most people something they're told to do. but for my father, who was a mathematician, it wasn't. most of us think of the problems in a math book as a way to test or develop our knowledge of the material explained in each section. but to my father the problems were the part that mattered, and the text was merely a sort of annotation. whenever he got a new math book it was to him like being given a puzzle: here was a new set of problems to solve, and he'd immediately set about solving all of them.

the other sense of a project being one's own — working on it by oneself — has a much softer edge. it shades gradually into collaboration. and interestingly, it shades into collaboration in two different ways. one way to collaborate is to share a single project. for example, when two mathematicians collaborate on a proof that takes shape in the course of a conversation between them. the other way is when multiple people work on separate projects of their own that fit together like a jigsaw puzzle. for example, when one person writes the text of a book and another does the graphic design. [3]

these two paths into collaboration can of course be combined. but under the right conditions, the excitement of working on a project of one's own can be preserved for quite a while before disintegrating into the turbulent flow of work in a large organization. indeed, the history of successful organizations is partly the history of techniques for preserving that excitement. [4]

the team that made the original macintosh were a great example of this phenomenon. people like burrell smith and andy hertzfeld and bill atkinson and susan kare were not just following orders. they were not tennis balls hit by steve jobs, but rockets let loose by steve jobs. there was a lot of collaboration between them, but they all seem to have individually felt the excitement of working on a project of one's own.

in andy hertzfeld's book on the macintosh, he describes how they'd come back into the office after dinner and work late into the night. people who've never experienced the thrill of working on a project they're excited about can't distinguish this kind of working long hours from the kind that happens in sweatshops and boiler rooms, but they're at opposite ends of the spectrum. that's why it's a mistake to insist dogmatically on "work/life balance." indeed, the mere expression "work/life" embodies a mistake: it assumes work and life are distinct. for those to whom the word "work" automatically implies the dutiful plodding kind, they are. but for the skaters, the relationship between work and life would be better represented by a dash than a slash. i wouldn't want to work on anything that i didn't want to take over my life.

of course, it's easier to achieve this level of motivation when you're making something like the macintosh. it's easy for something new to feel like a project of your own. that's one of the reasons for the tendency programmers have to rewrite things that don't need rewriting, and to write their own versions of things that already exist. this sometimes alarms managers, and measured by total number of

one metaphorically rode as a child rides a hobby-horse. it's hard to say if its recent, narrower meaning is a change for the better or the worse. for sure there are lots of false positives — lots of projects that end up being important but are dismissed initially as mere hobbies. but on the other hand, the concept provides valuable cover for projects in the early, ugly duckling phase.

2 tiger parents, as parents so often do, are fighting the last war. grades mattered more in the old days when the route to success was to acquire credentials while ascending some predefined ladder. but it's just as well that their tactics are focused on grades. how awful it would be if they invaded the territory of projects, and thereby gave their kids a distaste for this kind of work by forcing them to do it. grades are already a grim, fake world, and aren't harmed much by parental interference, but working on one's own projects is a more delicate, private thing that could be damaged very easily.

3 the complicated, gradual edge between working on one's own projects and collaborating with others is one reason there is so much disagreement about the idea of the "lone genius." in practice people collaborate (or not) in all kinds of different ways, but the idea of the lone genius is definitely not a myth. there's a core of truth to it that goes with a certain way of working.

4 collaboration is powerful too. the optimal organization would combine collaboration and ownership in such a way as to do the least damage to each. interestingly, companies and university departments approach this ideal from opposite directions: companies insist on collaboration, and occasionally also manage both to recruit skaters and allow them to skate, and university departments insist on the ability to do independent research (which is by custom treated as skating, whether it is or not), and the people they hire collaborate as much as they choose.

5 if a company could design its software in such a way that the best newly arrived programmers always got a clean sheet, it could have a kind of eternal youth. that might not be impossible. if you had a software backbone defining a game with sufficiently clear rules, individual programmers could write their own players.

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02

the bus ticket theory of genius

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the bus ticket theory is similar to carlyle's famous definition of genius as an infinite capacity for taking pains. but there are two differences. the bus ticket theory makes it clear that the source of this infinite capacity for taking pains is not infinite diligence, as carlyle seems to have meant, but the sort of infinite interest that collectors have. it also adds an important qualification: an infinite capacity for taking pains about something that matters.

so what matters? you can never be sure. it's precisely because no one can tell in advance which paths are promising that you can discover new ideas by working on what you're interested in.

but there are some heuristics you can use to guess whether an obsession might be one that matters. for example, it's more promising if you're creating something, rather than just consuming something someone else creates. it's more promising if something you're interested in is difficult, especially if it's more difficult for other people than it is for you. and the obsessions of talented people are more likely to be promising. when talented people become interested in random things, they're not truly random.

but you can never be sure. in fact, here's an interesting idea that's also rather alarming if it's true: it may be that to do great work, you also have to waste a lot of time.

in many different areas, reward is proportionate to risk. if that rule holds here, then the way to find paths that lead to truly great work is to be willing to expend a lot of effort on things that turn out to be every bit as unpromising as they seem.

i'm not sure if this is true. on one hand, it seems surprisingly difficult to waste your time so long as you're working hard on something interesting. so much of what you do ends up being useful. but on the other hand, the rule about the relationship between risk and reward is so powerful that it seems to hold wherever risk occurs. newton's case, at least, suggests that the risk/reward rule holds here. he's famous for one particular obsession of his that turned out to be unprecedentedly fruitful: using math to describe the world. but he had two other obsessions, alchemy and theology, that seem to have been complete wastes of time. he ended up net ahead. his bet on what we now call physics paid off so well that it more than compensated for the other two. but were the other two necessary, in the sense that he had to take big risks to make such big discoveries? i don't know.

here's an even more alarming idea: might one make all bad bets? it probably happens quite often. but we don't know how often, because these people don't become famous.

it's not merely that the returns from following a path are hard to predict. they change dramatically over time. 1830 was a really good time to be obsessively interested in natural history. if darwin had been born in 1709 instead of 1809, we might never have heard of him.

what can one do in the face of such uncertainty? one solution is to hedge your bets, which in this case means to follow the obviously promising paths instead of your own private obsessions. but as with any hedge, you're decreasing reward when you decrease risk. if you forgo working on what you like in order to follow some more conventionally ambitious path, you might miss something wonderful that you'd otherwise have discovered. that too must happen all the time, perhaps even more often than the genius whose bets all fail.

the other solution is to let yourself be interested in lots of different things. you don't decrease your upside if you switch between equally genuine interests based on which seems to be working so far. but there is a danger here too: if you work on too many different projects, you might not get deeply enough into any of them.

y still very young if we haven't nailed something so basic. it's exciting to think there are still discoveries to make about discovery. if that's the sort of thing you're interested in.

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notes

1 there are other types of collecting that illustrate this point better than bus tickets, but they're also more popular. it seemed just as well to use an inferior example rather than offend more people by telling them their hobby doesn't matter.

2 i worried a little about using the word "disinterested," since some people mistakenly believe it means not interested. but anyone who expects to be a genius will have to know the meaning of such a basic word, so i figure they may as well start now.

3 think how often genius must have been nipped in the bud by people being told, or telling themselves, to stop messing about and be responsible. ramanujan's mother was a huge enabler. imagine if she hadn't been. imagine if his parents had made him go out and get a job instead of sitting around at home doing math.

on the other hand, anyone quoting the preceding paragraph to justify not getting a job is probably mistaken.

4 1709 darwin is to time what the milanese leonardo is to space.

5 "an infinite capacity for taking pains" is a paraphrase of what carlyle wrote. what he wrote, in his *history of frederick the great*, was "... it is the fruit of 'genius' (which means transcendent capacity of taking trouble, first of all)...." since the paraphrase seems the name of the idea at this point, i kept it.

carlyle's *history* was published in 1858. in 1785 hérault de séchelles quoted buffon as saying "le génie n'est qu'une plus grande aptitude à la patience." (genius is only a greater aptitude for patience.)

6 trollope was establishing the system of postal routes. he himself sensed the obsessiveness with which he pursued this goal.

> it is amusing to watch how a passion will grow upon a man. during those two years it was the ambition of my life to cover the country with rural letter-carriers.

even newton occasionally sensed the degree of his obsessiveness. after computing pi to 15 digits, he wrote in a letter to a friend:

> i am ashamed to tell you to how many figures i carried these computations, having no other business at the time.

incidentally, ramanujan was also a compulsive calculator. as kanigel writes in his excellent biography:

> one ramanujan scholar, b. m. wilson, later told how ramanujan's research into number theory was often "preceded by a table of numerical results, carried usually to a length from which most of us would shrink."

[7] working to understand the natural world counts as creating rather than consuming.

newton tripped over this distinction when he chose to work on theology. his beliefs did not allow him to see it, but chasing down paradoxes in nature is fruitful in a way that chasing down paradoxes in sacred texts is not.

8 how much of people's propensity to become interested in a topic is inborn? my experience so far suggests the answer is: most of it. different kids get interested in different things, and it's hard to make a child interested in

03

how to think for yourself

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can you make yourself more independent-minded? i think so. this quality may be largely inborn, but there seem to be ways to magnify it, or at least not to suppress it.

one of the most effective techniques is one practiced unintentionally by most nerds: simply to be less aware what conventional beliefs are. it's hard to be a conformist if you don't know what you're supposed to conform to. though again, it may be that such people already are independent-minded. a conventional-minded person would probably feel anxious not knowing what other people thought, and make more effort to find out.

it matters a lot who you surround yourself with. if you're surrounded by conventional-minded people, it will constrain which ideas you can express, and that in turn will constrain which ideas you have. but if you surround yourself with independent-minded people, you'll have the opposite experience: hearing other people say surprising things will encourage you to, and to think of more.

because the independent-minded find it uncomfortable to be surrounded by conventional-minded people, they tend to self-segregate once they have a chance to. the problem with high school is that they haven't yet had a chance to. plus high school tends to be an inward-looking little world whose inhabitants lack confidence, both of which magnify the forces of conformism. so high school is often a bad time for the independent-minded. but there is some advantage even here: it teaches you what to avoid. if you later find yourself in a situation that makes you think "this is like high school," you know you should get out. [2]

another place where the independent- and conventional-minded are thrown together is in successful startups. the founders and early employees are almost always independent-minded; otherwise the startup wouldn't be successful. but conventional-minded people greatly outnumber independent-minded ones, so as the company grows, the original spirit of independent-mindedness is inevitably diluted. this causes all kinds of problems besides the obvious one that the company starts to suck. one of the strangest is that the founders find themselves able to speak more freely with founders of other companies than with their own employees. [3]

fortunately you don't have to spend all your time with independent-minded people. it's enough to have one or two you can talk to regularly. and once you find them, they're usually as eager to talk as you are; they need you too. although universities no longer have the kind of monopoly they used to have on education, good universities are still an excellent way to meet independent-minded people. most students will still be conventional-minded, but you'll at least find clumps of independent-minded ones, rather than the near zero you may have found in high school.

it also works to go in the other direction: as well as cultivating a small collection of independent-minded friends, to try to meet as many different types of people as you can. it will decrease the influence of your immediate peers if you have several other groups of peers. plus if you're part of several different worlds, you can often import ideas from one to another.

but by different types of people, i don't mean demographically different. for this technique to work, they have to think differently. so while it's an excellent idea to go and visit other countries, you can probably find people who think differently right around the corner. when i meet someone who knows a lot about something unusual (which includes practically everyone, if you dig deep enough), i try to learn what they know that other people don't. there are almost always surprises here. it's a good way to make conversation when you meet strangers, but i don't do it to make conversation. i really want to know.

been manufactured to capture them, they end up being less independent-minded than ordinary people, because they're subject to a much more exacting master than mere convention. [7]

can you increase your fastidiousness about truth? i would think so. in my experience, merely thinking about something you're fastidious about causes that fastidiousness to grow. if so, this is one of those rare virtues we can have more of merely by wanting it. and if it's like other forms of fastidiousness, it should also be possible to encourage in children. i certainly got a strong dose of it from my father. [8]

the second component of independent-mindedness, resistance to being told what to think, is the most visible of the three. but even this is often misunderstood. the big mistake people make about it is to think of it as a merely negative quality. the language we use reinforces that idea. you're *un*conventional. you *don't* care what other people think. but it's not just a kind of immunity. in the most independent-minded people, the desire not to be told what to think is a positive force. it's not mere skepticism, but an active delight in ideas that subvert the conventional wisdom, the more counterintuitive the better.

some of the most novel ideas seemed at the time almost like practical jokes. think how often your reaction to a novel idea is to laugh. i don't think it's because novel ideas are funny per se, but because novelty and humor share a certain kind of surprisingness. but while not identical, the two are close enough that there is a definite correlation between having a sense of humor and being independent-minded — just as there is between being humorless and being conventional-minded. [9]

i don't think we can significantly increase our resistance to being told what to think. it seems the most innate of the three components of independent-mindedness; people who have this quality as adults usually showed all too visible signs of it as children. but if we can't increase our resistance to being told what to think, we can at least shore it up, by surrounding ourselves with other independent-minded people.

the third component of independent-mindedness, curiosity, may be the most interesting. to the extent that we can give a brief answer to the question of where novel ideas come from, it's curiosity. that's what people are usually feeling before having them.

in my experience, independent-mindedness and curiosity predict one another perfectly. everyone i know who's independent-minded is deeply curious, and everyone i know who's conventional-minded isn't. except, curiously, children. all small children are curious. perhaps the reason is that even the conventional-minded have to be curious in the beginning, in order to learn what the conventions are. whereas the independent-minded are the gluttons of curiosity, who keep eating even after they're full. [10]

the three components of independent-mindedness work in concert: fastidiousness about truth and resistance to being told what to think leave space in your brain, and curiosity finds new ideas to fill it.

interestingly, the three components can substitute for one another in much the same way muscles can. if you're sufficiently fastidious about truth, you don't need to be as resistant to being told what to think, because fastidiousness alone will create sufficient gaps in your knowledge. and either one can compensate for curiosity, because if you create enough space in your brain, your discomfort at the resulting vacuum will add force to your curiosity. or curiosity can compensate for them: if you're sufficiently curious, you don't need to clear space in your brain, because the new ideas you discover will push out the conventional ones you acquired by default.

because the components of independent-mindedness are so interchangeable, you can have them to varying degrees and still get the same result. so there is not just a single model of independent-minde-

competency" should be augmented by one about outsourcing work done by people who'd ruin your culture as employees.

some investment firms already seem to be able to grow revenues without growing the number of employees. automation plus the ever increasing articulation of the "tech stack" suggest this may one day be possible for product companies.

4 there are intellectual fashions in every field, but their influence varies. one of the reasons politics, for example, tends to be boring is that it's so extremely subject to them. the threshold for having opinions about politics is much lower than the one for having opinions about set theory. so while there are some ideas in politics, in practice they tend to be swamped by waves of intellectual fashion.

5 the conventional-minded are often fooled by the strength of their opinions into believing that they're independent-minded. but strong convictions are not a sign of independent-mindedness. rather the opposite.

6 fastidiousness about truth doesn't imply that an independent-minded person won't be dishonest, but that he won't be deluded. it's sort of like the definition of a gentleman as someone who is never unintentionally rude.

7 you see this especially among political extremists. they think themselves nonconformists, but actually they're niche conformists. their opinions may be different from the average person's, but they are often more influenced by their peers' opinions than the average person's are.

8 if we broaden the concept of fastidiousness about truth so that it excludes pandering, bogusness, and pomposity as well as falsehood in the strict sense, our model of independent-mindedness can expand further into the arts.

9 this correlation is far from perfect, though. gödel and dirac don't seem to have been very strong in the humor department. but someone who is both "neurotypical" and humorless is very likely to be conventional-minded.

10 exception: gossip. almost everyone is curious about gossip.

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04

the lesson to unlearn

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sharply-defined chunks of information. if one of the assigned readings has an interesting digression on some subtle point, you can safely ignore that, because it's not the sort of thing that could be turned into a test question. but if the professor tells you that there were three underlying causes of the schism of 1378, or three main consequences of the black death, you'd better know them. and whether they were in fact the causes or consequences is beside the point. for the purposes of this class they are.

at a university there are often copies of old exams floating around, and these narrow still further what you have to learn. as well as learning what kind of questions this professor asks, you'll often get actual exam questions. many professors re-use them. after teaching a class for 10 years, it would be hard not to, at least inadvertently.

in some classes, your professor will have had some sort of political axe to grind, and if so you'll have to grind it too. the need for this varies. in classes in math or the hard sciences or engineering it's rarely necessary, but at the other end of the spectrum there are classes where you couldn't get a good grade without it.

getting a good grade in a class on x is so different from learning a lot about x that you have to choose one or the other, and you can't blame students if they choose grades. everyone judges them by their grades — graduate programs, employers, scholarships, even their own parents.

i liked learning, and i really enjoyed some of the papers and programs i wrote in college. but did i ever, after turning in a paper in some class, sit down and write another just for fun? of course not. i had things due in other classes. if it ever came to a choice of learning or grades, i chose grades. i hadn't come to college to do badly.

anyone who cares about getting good grades has to play this game, or they'll be surpassed by those who do. and at elite universities, that means nearly everyone, since someone who didn't care about getting good grades probably wouldn't be there in the first place. the result is that students compete to maximize the difference between learning and getting good grades.

why are tests so bad? more precisely, why are they so hackable? any experienced programmer could answer that. how hackable is software whose author hasn't paid any attention to preventing it from being hacked? usually it's as porous as a colander.

hackable is the default for any test imposed by an authority. the reason the tests you're given are so consistently bad — so consistently far from measuring what they're supposed to measure — is simply that the people creating them haven't made much effort to prevent them from being hacked.

but you can't blame teachers if their tests are hackable. their job is to teach, not to create unhackable tests. the real problem is grades, or more precisely, that grades have been overloaded. if grades were merely a way for teachers to tell students what they were doing right and wrong, like a coach giving advice to an athlete, students wouldn't be tempted to hack tests. but unfortunately after a certain age grades become more than advice. after a certain age, whenever you're being taught, you're usually also being judged.

i've used college tests as an example, but those are actually the least hackable. all the tests most students take their whole lives are at least as bad, including, most spectacularly of all, the test that gets them into college. if getting into college were merely a matter of having the quality of one's mind measured by admissions officers the way scientists measure the mass of an object, we could tell teenage kids "learn a lot" and leave it at that. you can tell how bad college admissions are, as a test, from how unlike high school that sounds. in practice, the freakishly specific nature of the stuff ambitious kids have to do in high school is directly proportionate to the hackability of college admissions. the classes

it would sometimes take founders years to grasp these simple lessons. and not because they were lazy or stupid. they just seemed blind to what was right in front of them.

why, i would ask myself, do they always make things so complicated? and then one day i realized this was not a rhetorical question.

why did founders tie themselves in knots doing the wrong things when the answer was right in front of them? because that was what they'd been trained to do. their education had taught them that the way to win was to hack the test. and without even telling them they were being trained to do this. the younger ones, the recent graduates, had never faced a non-artificial test. they thought this was just how the world worked: that the first thing you did, when facing any kind of challenge, was to figure out what the trick was for hacking the test. that's why the conversation would always start with how to raise money, because that read as the test. it came at the end of yc. it had numbers attached to it, and higher numbers seemed to be better. it must be the test.

there are certainly big chunks of the world where the way to win is to hack the test. this phenomenon isn't limited to schools. and some people, either due to ideology or ignorance, claim that this is true of startups too. but it isn't. in fact, one of the most striking things about startups is the degree to which you win by simply doing good work. there are edge cases, as there are in anything, but in general you win by getting users, and what users care about is whether the product does what they want.

why did it take me so long to understand why founders made startups overcomplicated? because i hadn't realized explicitly that schools train us to win by hacking bad tests. and not just them, but me! i'd been trained to hack bad tests too, and hadn't realized it till decades later.

i had lived as if i realized it, but without knowing why. for example, i had avoided working for big companies. but if you'd asked why, i'd have said it was because they were bogus, or bureaucratic. or just yuck. i never understood how much of my dislike of big companies was due to the fact that you win by hacking bad tests.

similarly, the fact that the tests were unhackable was a lot of what attracted me to startups. but again, i hadn't realized that explicitly.

i had in effect achieved by successive approximations something that may have a closed-form solution. i had gradually undone my training in hacking bad tests without knowing i was doing it. could someone coming out of school banish this demon just by knowing its name, and saying begone? it seems worth trying.

merely talking explicitly about this phenomenon is likely to make things better, because much of its power comes from the fact that we take it for granted. after you've noticed it, it seems the elephant in the room, but it's a pretty well camouflaged elephant. the phenomenon is so old, and so pervasive. and it's simply the result of neglect. no one meant things to be this way. this is just what happens when you combine learning with grades, competition, and the naive assumption of unhackability.

it was mind-blowing to realize that two of the things i'd puzzled about the most — the bogusness of high school, and the difficulty of getting founders to see the obvious — both had the same cause. it's rare for such a big block to slide into place so late.

usually when that happens it has implications in a lot of different areas, and this case seems no exception. for example, it suggests both that education could be done better, and how you might fix it. but it also suggests a potential answer to the question all big companies seem to have: how can we be more

notes

1 if using tests only to measure learning sounds impossibly utopian, that is already the way things work at lambda school. lambda school doesn't have grades. you either graduate or you don't. the only purpose of tests is to decide at each stage of the curriculum whether you can continue to the next. so in effect the whole school is pass/fail.

2 if the final exam consisted of a long conversation with the professor, you could prepare for it by reading good books on medieval history. a lot of the hackability of tests in schools is due to the fact that the same test has to be given to large numbers of students.

3 learning is the naive algorithm for getting good grades.

4 hacking has multiple senses. there's a narrow sense in which it means to compromise something. that's the sense in which one hacks a bad test. but there's another, more general sense, meaning to find a surprising solution to a problem, often by thinking differently about it. hacking in this sense is a wonderful thing. and indeed, some of the hacks people use on bad tests are impressively ingenious; the problem is not so much the hacking as that, because the tests are hackable, they don't test what they're meant to.

5 the people who pick startups at y combinator are similar to admissions officers, except that instead of being arbitrary, their acceptance criteria are trained by a very tight feedback loop. if you accept a bad startup or reject a good one, you will usually know it within a year or two at the latest, and often within a month.

6 i'm sure admissions officers are tired of reading applications from kids who seem to have no personality beyond being willing to seem however they're supposed to seem to get accepted. what they don't realize is that they are, in a sense, looking in a mirror. the lack of authenticity in the applicants is a reflection of the arbitrariness of the application process. a dictator might just as well complain about the lack of authenticity in the people around him.

7 by good work, i don't mean morally good, but good in the sense in which a good craftsman does good work.

8 there are borderline cases where it's hard to say which category a test falls in. for example, is raising venture capital like college admissions, or is it like selling to a customer?

9 note that a good test is merely one that's unhackable. good here doesn't mean morally good, but good in the sense of working well. the difference between fields with bad tests and good ones is not that the former are bad and the latter are good, but that the former are bogus and the latter aren't. but those two measures are not unrelated. as tara ploughman said, the path from good to evil goes through bogus.

10 people who think the recent increase in economic inequality is due to changes in tax policy seem very naive to anyone with experience in startups. different people are getting rich now than used to, and they're getting much richer than mere tax savings could make them.

11 note to tiger parents: you may think you're training your kids to win, but if you're training them to win by hacking bad tests, you are, as parents so often do, training them to fight the last war.

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05

how to work hard

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there wasn't a single point when i learned this. like most little kids, i enjoyed the feeling of achievement when i learned or did something new. as i grew older, this morphed into a feeling of disgust when i wasn't achieving anything. the one precisely dateable landmark i have is when i stopped watching tv, at age 13.

several people i've talked to remember getting serious about work around this age. when i asked patrick collison when he started to find idleness distasteful, he said

> i think around age 13 or 14. i have a clear memory from around then of sitting in the sitting room, staring outside, and wondering why i was wasting my summer holiday.

perhaps something changes at adolescence. that would make sense.

strangely enough, the biggest obstacle to getting serious about work was probably school, which made work (what they called work) seem boring and pointless. i had to learn what real work was before i could wholeheartedly desire to do it. that took a while, because even in college a lot of the work is pointless; there are entire departments that are pointless. but as i learned the shape of real work, i found that my desire to do it slotted into it as if they'd been made for each other.

i suspect most people have to learn what work is before they can love it. hardy wrote eloquently about this in *a mathematician's apology*:

> i do not remember having felt, as a boy, any *passion* for mathematics, and such notions as i may have had of the career of a mathematician were far from noble. i thought of mathematics in terms of examinations and scholarships: i wanted to beat other boys, and this seemed to be the way in which i could do so most decisively.

he didn't learn what math was really about till part way through college, when he read jordan's *cours d'analyse*.

> i shall never forget the astonishment with which i read that remarkable work, the first inspiration for so many mathematicians of my generation, and learnt for the first time as i read it what mathematics really meant.

there are two separate kinds of fakeness you need to learn to discount in order to understand what real work is. one is the kind hardy encountered in school. subjects get distorted when they're adapted to be taught to kids — often so distorted that they're nothing like the work done by actual practitioners. [3] the other kind of fakeness is intrinsic to certain types of work. some types of work are inherently bogus, or at best mere busywork.

there's a kind of solidity to real work. it's not all writing the *principia*, but it all feels necessary. that's a vague criterion, but it's deliberately vague, because it has to cover a lot of different types. [4]

once you know the shape of real work, you have to learn how many hours a day to spend on it. you can't solve this problem by simply working every waking hour, because in many kinds of work there's a point beyond which the quality of the result will start to decline.

that limit varies depending on the type of work and the person. i've done several different kinds of work, and the limits were different for each. my limit for the harder types of writing or programming is about five hours a day. whereas when i was running a startup, i could work all the time. at least for the three years i did it; if i'd kept going much longer, i'd probably have needed to take occasional vacations. [5]

it can be harder to discover your interests than your talents. there are fewer types of talent than interest, and they start to be judged early in childhood, whereas interest in a topic is a subtle thing that may not mature till your twenties, or even later. the topic may not even exist earlier. plus there are some powerful sources of error you need to learn to discount. are you really interested in x, or do you want to work on it because you'll make a lot of money, or because other people will be impressed with you, or because your parents want you to? [8]

the difficulty of figuring out what to work on varies enormously from one person to another. that's one of the most important things i've learned about work since i was a kid. as a kid, you get the impression that everyone has a calling, and all they have to do is figure out what it is. that's how it works in movies, and in the streamlined biographies fed to kids. sometimes it works that way in real life. some people figure out what to do as children and just do it, like mozart. but others, like newton, turn restlessly from one kind of work to another. maybe in retrospect we can identify one as their calling — we can wish newton spent more time on math and physics and less on alchemy and theology — but this is an illusion induced by hindsight bias. there was no voice calling to him that he could have heard.

so while some people's lives converge fast, there will be others whose lives never converge. and for these people, figuring out what to work on is not so much a prelude to working hard as an ongoing part of it, like one of a set of simultaneous equations. for these people, the process i described earlier has a third component: along with measuring both how hard you're working and how well you're doing, you have to think about whether you should keep working in this field or switch to another. if you're working hard but not getting good enough results, you should switch. it sounds simple expressed that way, but in practice it's very difficult. you shouldn't give up on the first day just because you work hard and don't get anywhere. you need to give yourself time to get going. but how much time? and what should you do if work that was going well stops going well? how much time do you give yourself then? [9]

what even counts as good results? that can be really hard to decide. if you're exploring an area few others have worked in, you may not even know what good results look like. history is full of examples of people who misjudged the importance of what they were working on.

the best test of whether it's worthwhile to work on something is whether you find it interesting. that may sound like a dangerously subjective measure, but it's probably the most accurate one you're going to get. you're the one working on the stuff. who's in a better position than you to judge whether it's important, and what's a better predictor of its importance than whether it's interesting?

for this test to work, though, you have to be honest with yourself. indeed, that's the most striking thing about the whole question of working hard: how at each point it depends on being honest with yourself.

working hard is not just a dial you turn up to 11. it's a complicated, dynamic system that has to be tuned just right at each point. you have to understand the shape of real work, see clearly what kind you're best suited for, aim as close to the true core of it as you can, accurately judge at each moment both what you're capable of and how you're doing, and put in as many hours each day as you can without harming the quality of the result. this network is too complicated to trick. but if you're consistently honest and clear-sighted, it will automatically assume an optimal shape, and you'll be productive in a way few people are.

06

the right kind of stubborn

july 2024
paulgraham.com/persistence.html

them solve harder problems. but that's not what happens. if you handed the collisons an extremely hard problem to solve, they wouldn't become obstinate. if anything they'd become less obstinate. they'd know they had to be open to anything.

similarly, if obstinacy were caused by the situation, the obstinate would stop being obstinate when solving easier problems. but they don't. and if obstinacy isn't caused by the situation, it must come from within. it must be a feature of one's personality.

obstinacy is a reflexive resistance to changing one's ideas. this is not identical with stupidity, but they're closely related. a reflexive resistance to changing one's ideas becomes a sort of induced stupidity as contrary evidence mounts. and obstinacy is a form of not giving up that's easily practiced by the stupid. you don't have to consider complicated tradeoffs; you just dig in your heels. it even works, up to a point.

the fact that obstinacy works for simple problems is an important clue. persistence and obstinacy aren't opposites. the relationship between them is more like the relationship between the two kinds of respiration we can do: aerobic respiration, and the anaerobic respiration we inherited from our most distant ancestors. anaerobic respiration is a more primitive process, but it has its uses. when you leap suddenly away from a threat, that's what you're using.

the optimal amount of obstinacy is not zero. it can be good if your initial reaction to a setback is an unthinking "i won't give up," because this helps prevent panic. but unthinking only gets you so far. the further someone is toward the obstinate end of the continuum, the less likely they are to succeed in solving hard problems. [4]

obstinacy is a simple thing. animals have it. but persistence turns out to have a fairly complicated internal structure.

one thing that distinguishes the persistent is their energy. at the risk of putting too much weight on words, they persist rather than merely resisting. they keep trying things. which means the persistent must also be imaginative. to keep trying things, you have to keep thinking of things to try.

energy and imagination make a wonderful combination. each gets the best out of the other. energy creates demand for the ideas produced by imagination, which thus produces more, and imagination gives energy somewhere to go. [5]

merely having energy and imagination is quite rare. but to solve hard problems you need three more qualities: resilience, good judgement, and a focus on some kind of goal.

resilience means not having one's morale destroyed by setbacks. setbacks are inevitable once problems reach a certain size, so if you can't bounce back from them, you can only do good work on a small scale. but resilience is not the same as obstinacy. resilience means setbacks can't change your morale, not that they can't change your mind.

indeed, persistence often requires that one change one's mind. that's where good judgement comes in. the persistent are quite rational. they focus on expected value. it's this, not recklessness, that lets them work on things that are unlikely to succeed.

there is one point at which the persistent are often irrational though: at the very top of the decision tree. when they choose between two problems of roughly equal expected value, the choice usually comes down to personal preference. indeed, they'll often classify projects into deliberately wide bands of expected value in order to ensure that the one they want to work on still qualifies.

6 start by erring on the small side. if you're inexperienced you'll inevitably err on one side or the other, and if you err on the side of making the goal too broad, you won't get anywhere. whereas if you err on the small side you'll at least be moving forward. then, once you're moving, you expand the goal.

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07

what to do

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paulgraham.com/do.html

i'm not too worried that the traditional answers don't mention taking care of the world. obviously people only started to care about that once it became clear we could ruin it. but how can making good new things be important if the traditional answers don't mention it?

the traditional answers were answers to a slightly different question. they were answers to the question of how to be, rather than what to do. the audience didn't have a lot of choice about what to do. the audience up till recent centuries was the landowning class, which was also the political class. they weren't choosing between doing physics and writing novels. their work was foreordained: manage their estates, participate in politics, fight when necessary. it was ok to do certain other kinds of work in one's spare time, but ideally one didn't have any. cicero's *de officiis* is one of the great classical answers to the question of how to live, and in it he explicitly says that he wouldn't even be writing it if he hadn't been excluded from public life by recent political upheavals. [2]

there were of course people doing what we would now call "original work," and they were often admired for it, but they weren't seen as models. archimedes knew that he was the first to prove that a sphere has 2/3 the volume of the smallest enclosing cylinder and was very pleased about it. but you don't find ancient writers urging their readers to emulate him. they regarded him more as a prodigy than a model.

now many more of us can follow archimedes's example and devote most of our attention to one kind of work. he turned out to be a model after all, along with a collection of other people that his contemporaries would have found it strange to treat as a distinct group, because the vein of people making new things ran at right angles to the social hierarchy.

what kinds of new things count? i'd rather leave that question to the makers of them. it would be a risky business to try to define any kind of threshold, because new kinds of work are often despised at first. raymond chandler was writing literal pulp fiction, and he's now recognized as one of the best writers of the twentieth century. indeed this pattern is so common that you can use it as a recipe: if you're excited about some kind of work that's not considered prestigious and you can explain what everyone else is overlooking about it, then this is not merely a kind of work that's ok to do, but one to seek out.

the other reason i wouldn't want to define any thresholds is that we don't need them. the kind of people who make good new things don't need rules to keep them honest.

so there's my guess at a set of principles to live by: take care of people and the world, and make good new things. different people will do these to varying degrees. there will presumably be lots who focus entirely on taking care of people. there will be a few who focus mostly on making new things. but even if you're one of those, you should at least make sure that the new things you make don't net *harm* people or the world. and if you go a step further and try to make things that help them, you may find you're ahead on the trade. you'll be more constrained in what you can make, but you'll make it with more energy.

on the other hand, if you make something amazing, you'll often be helping people or the world even if you didn't mean to. newton was driven by curiosity and ambition, not by any practical effect his work might have, and yet the practical effect of his work has been enormous. and this seems the rule rather than the exception. so if you think you can make something amazing, you should probably just go ahead and do it.

08

when to do what you love

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paulgraham.com/when.html

when you can't decide which path to take, it's almost always due to ignorance. in fact you're usually suffering from three kinds of ignorance simultaneously: you don't know what makes you happy, what the various kinds of work are really like, or how well you could do them. [2]

in a way this ignorance is excusable. it's often hard to predict these things, and no one even tells you that you need to. if you're ambitious you're told you should go to college, and this is good advice so far as it goes, but that's where it usually ends. no one tells you how to figure out what to work on, or how hard this can be.

what do you do in the face of uncertainty? get more certainty. and probably the best way to do that is to try working on things you're interested in. that will get you more information about how interested you are in them, how good you are at them, and how much scope they offer for ambition.

don't wait. don't wait till the end of college to figure out what to work on. don't even wait for internships during college. you don't necessarily need a job doing x in order to work on x; often you can just start doing it in some form yourself. and since figuring out what to work on is a problem that could take years to solve, the sooner you start, the better.

one useful trick for judging different kinds of work is to look at who your colleagues will be. you'll become like whoever you work with. do you want to become like these people?

indeed, the difference in character between different kinds of work is magnified by the fact that everyone else is facing the same decisions as you. if you choose a kind of work mainly for how well it pays, you'll be surrounded by other people who chose it for the same reason, and that will make it even more soul-sucking than it seems from the outside. whereas if you choose work you're genuinely interested in, you'll be surrounded mostly by other people who are genuinely interested in it, and that will make it extra inspiring. [3]

the other thing you do in the face of uncertainty is to make choices that are uncertainty-proof. the less sure you are about what to do, the more important it is to choose options that give you more options in the future. i call this "staying upwind." if you're unsure whether to major in math or economics, for example, choose math; math is upwind of economics in the sense that it will be easier to switch later from math to economics than from economics to math.

there's one case, though, where it's easy to say whether you should work on what interests you the most: if you want to do great work. this is not a sufficient condition for doing great work, but it is a necessary one.

there's a lot of selection bias in advice about whether to "follow your passion," and this is the reason. most such advice comes from people who are famously successful, and if you ask someone who's famously successful how to do what they did, most will tell you that you have to work on what you're most interested in. and this is in fact true.

that doesn't mean it's the right advice for everyone. not everyone can do great work, or wants to. but if you do want to, the complicated question of whether or not to work on what interests you the most becomes simple. the answer is yes. the root of great work is a sort of ambitious curiosity, and you can't manufacture that.

09

life is short

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live in. counterintuitive as it feels, it's better most of the time not to defend yourself. otherwise these people are literally taking your life. [2]

arguing online is only incidentally addictive. there are more dangerous things than that. as i've written before, one byproduct of technical progress is that things we like tend to become more addictive. which means we will increasingly have to make a conscious effort to avoid addictions — to stand outside ourselves and ask "is this how i want to be spending my time?"

as well as avoiding bullshit, one should actively seek out things that matter. but different things matter to different people, and most have to learn what matters to them. a few are lucky and realize early on that they love math or taking care of animals or writing, and then figure out a way to spend a lot of time doing it. but most people start out with a life that's a mix of things that matter and things that don't, and only gradually learn to distinguish between them.

for the young especially, much of this confusion is induced by the artificial situations they find themselves in. in middle school and high school, what the other kids think of you seems the most important thing in the world. but when you ask adults what they got wrong at that age, nearly all say they cared too much what other kids thought of them.

one heuristic for distinguishing stuff that matters is to ask yourself whether you'll care about it in the future. fake stuff that matters usually has a sharp peak of seeming to matter. that's how it tricks you. the area under the curve is small, but its shape jabs into your consciousness like a pin.

the things that matter aren't necessarily the ones people would call "important." having coffee with a friend matters. you won't feel later like that was a waste of time.

one great thing about having small children is that they make you spend time on things that matter: them. they grab your sleeve as you're staring at your phone and say "will you play with me?" and odds are that is in fact the bullshit-minimizing option.

if life is short, we should expect its shortness to take us by surprise. and that is just what tends to happen. you take things for granted, and then they're gone. you think you can always write that book, or climb that mountain, or whatever, and then you realize the window has closed. the saddest windows close when other people die. their lives are short too. after my mother died, i wished i'd spent more time with her. i lived as if she'd always be there. and in her typical quiet way she encouraged that illusion. but an illusion it was. i think a lot of people make the same mistake i did.

the usual way to avoid being taken by surprise by something is to be consciously aware of it. back when life was more precarious, people used to be aware of death to a degree that would now seem a bit morbid. i'm not sure why, but it doesn't seem the right answer to be constantly reminding oneself of the grim reaper hovering at everyone's shoulder. perhaps a better solution is to look at the problem from the other end. cultivate a habit of impatience about the things you most want to do. don't wait before climbing that mountain or writing that book or visiting your mother. you don't need to be constantly reminding yourself why you shouldn't wait. just don't wait.

i can think of two more things one does when one doesn't have much of something: try to get more of it, and savor what one has. both make sense here.

how you live affects how long you live. most people could do better. me among them.

but you can probably get even more effect by paying closer attention to the time you have. it's easy to let the days rush by. the "flow" that imaginative people love so much has a darker cousin that prevents

10

how to do great work

july 2023
paulgraham.com/greatwork.html

ise to say precisely what they're all overlooking, that's as good a bet as you'll find. [3]

four steps: choose a field, learn enough to get to the frontier, notice gaps, explore promising ones. this is how practically everyone who's done great work has done it, from painters to physicists.

steps two and four will require hard work. it may not be possible to prove that you have to work hard to do great things, but the empirical evidence is on the scale of the evidence for mortality. that's why it's essential to work on something you're deeply interested in. interest will drive you to work harder than mere diligence ever could.

the three most powerful motives are curiosity, delight, and the desire to do something impressive. sometimes they converge, and that combination is the most powerful of all.

the big prize is to discover a new fractal bud. you notice a crack in the surface of knowledge, pry it open, and there's a whole world inside.

let's talk a little more about the complicated business of figuring out what to work on. the main reason it's hard is that you can't tell what most kinds of work are like except by doing them. which means the four steps overlap: you may have to work at something for years before you know how much you like it or how good you are at it. and in the meantime you're not doing, and thus not learning about, most other kinds of work. so in the worst case you choose late based on very incomplete information. [4]

the nature of ambition exacerbates this problem. ambition comes in two forms, one that precedes interest in the subject and one that grows out of it. most people who do great work have a mix, and the more you have of the former, the harder it will be to decide what to do.

the educational systems in most countries pretend it's easy. they expect you to commit to a field long before you could know what it's really like. and as a result an ambitious person on an optimal trajectory will often read to the system as an instance of breakage.

it would be better if they at least admitted it — if they admitted that the system not only can't do much to help you figure out what to work on, but is designed on the assumption that you'll somehow magically guess as a teenager. they don't tell you, but i will: when it comes to figuring out what to work on, you're on your own. some people get lucky and do guess correctly, but the rest will find themselves scrambling diagonally across tracks laid down on the assumption that everyone does.

what should you do if you're young and ambitious but don't know what to work on? what you should *not* do is drift along passively, assuming the problem will solve itself. you need to take action. but there is no systematic procedure you can follow. when you read biographies of people who've done great work, it's remarkable how much luck is involved. they discover what to work on as a result of a chance meeting, or by reading a book they happen to pick up. so you need to make yourself a big target for luck, and the way to do that is to be curious. try lots of things, meet lots of people, read lots of books, ask lots of questions. [5]

when in doubt, optimize for interestingness. fields change as you learn more about them. what mathematicians do, for example, is very different from what you do in high school math classes. so you need to give different types of work a chance to show you what they're like. but a field should become *increasingly* interesting as you learn more about it. if it doesn't, it's probably not for you.

don't worry if you find you're interested in different things than other people. the stranger your tastes in interestingness, the better. strange tastes are often strong ones, and a strong taste for work means you'll be productive. and you're more likely to find new things if you're looking where few have looked before.

in the sense that it's higher than the energy required to keep going, it's ok to tell yourself a lie of corresponding magnitude to get over it.

it's usually a mistake to lie to yourself if you want to do great work, but this is one of the rare cases where it isn't. when i'm reluctant to start work in the morning, i often trick myself by saying "i'll just read over what i've got so far." five minutes later i've found something that seems mistaken or incomplete, and i'm off.

similar techniques work for starting new projects. it's ok to lie to yourself about how much work a project will entail, for example. lots of great things began with someone saying "how hard could it be?"

this is one case where the young have an advantage. they're more optimistic, and even though one of the sources of their optimism is ignorance, in this case ignorance can sometimes beat knowledge.

try to finish what you start, though, even if it turns out to be more work than you expected. finishing things is not just an exercise in tidiness or self-discipline. in many projects a lot of the best work happens in what was meant to be the final stage.

another permissible lie is to exaggerate the importance of what you're working on, at least in your own mind. if that helps you discover something new, it may turn out not to have been a lie after all. [7]

since there are two senses of starting work — per day and per project — there are also two forms of procrastination. per-project procrastination is far the more dangerous. you put off starting that ambitious project from year to year because the time isn't quite right. when you're procrastinating in units of years, you can get a lot not done. [8]

one reason per-project procrastination is so dangerous is that it usually camouflages itself as work. you're not just sitting around doing nothing; you're working industriously on something else. so per-project procrastination doesn't set off the alarms that per-day procrastination does. you're too busy to notice it.

the way to beat it is to stop occasionally and ask yourself: am i working on what i most want to work on? when you're young it's ok if the answer is sometimes no, but this gets increasingly dangerous as you get older. [9]

great work usually entails spending what would seem to most people an unreasonable amount of time on a problem. you can't think of this time as a cost, or it will seem too high. you have to find the work sufficiently engaging as it's happening.

there may be some jobs where you have to work diligently for years at things you hate before you get to the good part, but this is not how great work happens. great work happens by focusing consistently on something you're genuinely interested in. when you pause to take stock, you're surprised how far you've come.

the reason we're surprised is that we underestimate the cumulative effect of work. writing a page a day doesn't sound like much, but if you do it every day you'll write a book a year. that's the key: consistency. people who do great things don't get a lot done every day. they get something done, rather than nothing.

if you do work that compounds, you'll get exponential growth. most people who do this do it unconsciously, but it's worth stopping to think about. learning, for example, is an instance of this phenomenon: the more you learn about something, the easier it is to learn more. growing an audience is another: the more fans you have, the more new fans they'll bring you.

the temptation to be someone else is greatest for the young. they often feel like nobodies. but you never need to worry about that problem, because it's self-solving if you work on sufficiently ambitious projects. if you succeed at an ambitious project, you're not a nobody; you're the person who did it. so just do the work and your identity will take care of itself.

"avoid affectation" is a useful rule so far as it goes, but how would you express this idea positively? how would you say what to be, instead of what not to be? the best answer is earnest. if you're earnest you avoid not just affectation but a whole set of similar vices.

the core of being earnest is being intellectually honest. we're taught as children to be honest as an unselfish virtue — as a kind of sacrifice. but in fact it's a source of power too. to see new ideas, you need an exceptionally sharp eye for the truth. you're trying to see more truth than others have seen so far. and how can you have a sharp eye for the truth if you're intellectually dishonest?

one way to avoid intellectual dishonesty is to maintain a slight positive pressure in the opposite direction. be aggressively willing to admit that you're mistaken. once you've admitted you were mistaken about something, you're free. till then you have to carry it. [13]

another more subtle component of earnestness is informality. informality is much more important than its grammatically negative name implies. it's not merely the absence of something. it means focusing on what matters instead of what doesn't.

what formality and affectation have in common is that as well as doing the work, you're trying to seem a certain way as you're doing it. but any energy that goes into how you seem comes out of being good. that's one reason nerds have an advantage in doing great work: they expend little effort on seeming anything. in fact that's basically the definition of a nerd.

nerds have a kind of innocent boldness that's exactly what you need in doing great work. it's not learned; it's preserved from childhood. so hold onto it. be the one who puts things out there rather than the one who sits back and offers sophisticated-sounding criticisms of them. "it's easy to criticize" is true in the most literal sense, and the route to great work is never easy.

there may be some jobs where it's an advantage to be cynical and pessimistic, but if you want to do great work it's an advantage to be optimistic, even though that means you'll risk looking like a fool sometimes. there's an old tradition of doing the opposite. the old testament says it's better to keep quiet lest you look like a fool. but that's advice for *seeming* smart. if you actually want to discover new things, it's better to take the risk of telling people your ideas.

some people are naturally earnest, and with others it takes a conscious effort. either kind of earnestness will suffice. but i doubt it would be possible to do great work without being earnest. it's so hard to do even if you are. you don't have enough margin for error to accommodate the distortions introduced by being affected, intellectually dishonest, orthodox, fashionable, or cool. [14]

great work is consistent not only with who did it, but with itself. it's usually all of a piece. so if you face a decision in the middle of working on something, ask which choice is more consistent.

you may have to throw things away and redo them. you won't necessarily have to, but you have to be willing to. and that can take some effort; when there's something you need to redo, status quo bias and laziness will combine to keep you in denial about it. to beat this ask: if i'd already made the change, would i want to revert to what i have now?

have the confidence to cut. don't keep something that doesn't fit just because you're proud of it, or because it cost you a lot of effort.

working on something. original ideas don't come from trying to have original ideas. they come from trying to build or understand something slightly too difficult. [15]

talking or writing about the things you're interested in is a good way to generate new ideas. when you try to put ideas into words, a missing idea creates a sort of vacuum that draws it out of you. indeed, there's a kind of thinking that can only be done by writing.

changing your context can help. if you visit a new place, you'll often find you have new ideas there. the journey itself often dislodges them. but you may not have to go far to get this benefit. sometimes it's enough just to go for a walk. [16]

it also helps to travel in topic space. you'll have more new ideas if you explore lots of different topics, partly because it gives the angle grinder more surface area to work on, and partly because analogies are an especially fruitful source of new ideas.

don't divide your attention *evenly* between many topics though, or you'll spread yourself too thin. you want to distribute it according to something more like a power law. [17] be professionally curious about a few topics and idly curious about many more.

curiosity and originality are closely related. curiosity feeds originality by giving it new things to work on. but the relationship is closer than that. curiosity is itself a kind of originality; it's roughly to questions what originality is to answers. and since questions at their best are a big component of answers, curiosity at its best is a creative force.

having new ideas is a strange game, because it usually consists of seeing things that were right under your nose. once you've seen a new idea, it tends to seem obvious. why did no one think of this before?

when an idea seems simultaneously novel and obvious, it's probably a good one.

seeing something obvious sounds easy. and yet empirically having new ideas is hard. what's the source of this apparent contradiction? it's that seeing the new idea usually requires you to change the way you look at the world. we see the world through models that both help and constrain us. when you fix a broken model, new ideas become obvious. but noticing and fixing a broken model is hard. that's how new ideas can be both obvious and yet hard to discover: they're easy to see after you do something hard.

one way to discover broken models is to be stricter than other people. broken models of the world leave a trail of clues where they bash against reality. most people don't want to see these clues. it would be an understatement to say that they're attached to their current model; it's what they think in; so they'll tend to ignore the trail of clues left by its breakage, however conspicuous it may seem in retrospect.

to find new ideas you have to seize on signs of breakage instead of looking away. that's what einstein did. he was able to see the wild implications of maxwell's equations not so much because he was looking for new ideas as because he was stricter.

the other thing you need is a willingness to break rules. paradoxical as it sounds, if you want to fix your model of the world, it helps to be the sort of person who's comfortable breaking rules. from the point of view of the old model, which everyone including you initially shares, the new model usually breaks at least implicit rules.

few understand the degree of rule-breaking required, because new ideas seem much more conservative once they succeed. they seem perfectly reasonable once you're using the new model of the world they

one of the most interesting kinds of unfashionable problem is the problem that people think has been fully explored, but hasn't. great work often takes something that already exists and shows its latent potential. durer and watt both did this. so if you're interested in a field that others think is tapped out, don't let their skepticism deter you. people are often wrong about this.

working on an unfashionable problem can be very pleasing. there's no hype or hurry. opportunists and critics are both occupied elsewhere. the existing work often has an old-school solidity. and there's a satisfying sense of economy in cultivating ideas that would otherwise be wasted.

but the most common type of overlooked problem is not explicitly unfashionable in the sense of being out of fashion. it just doesn't seem to matter as much as it actually does. how do you find these? by being self-indulgent — by letting your curiosity have its way, and tuning out, at least temporarily, the little voice in your head that says you should only be working on "important" problems.

you do need to work on important problems, but almost everyone is too conservative about what counts as one. and if there's an important but overlooked problem in your neighborhood, it's probably already on your subconscious radar screen. so try asking yourself: if you were going to take a break from "serious" work to work on something just because it would be really interesting, what would you do? the answer is probably more important than it seems.

originality in choosing problems seems to matter even more than originality in solving them. that's what distinguishes the people who discover whole new fields. so what might seem to be merely the initial step — deciding what to work on — is in a sense the key to the whole game.

few grasp this. one of the biggest misconceptions about new ideas is about the ratio of question to answer in their composition. people think big ideas are answers, but often the real insight was in the question.

part of the reason we underrate questions is the way they're used in schools. in schools they tend to exist only briefly before being answered, like unstable particles. but a really good question can be much more than that. a really good question is a partial discovery. how do new species arise? is the force that makes objects fall to earth the same as the one that keeps planets in their orbits? by even asking such questions you were already in excitingly novel territory.

unanswered questions can be uncomfortable things to carry around with you. but the more you're carrying, the greater the chance of noticing a solution — or perhaps even more excitingly, noticing that two unanswered questions are the same.

sometimes you carry a question for a long time. great work often comes from returning to a question you first noticed years before — in your childhood, even — and couldn't stop thinking about. people talk a lot about the importance of keeping your youthful dreams alive, but it's just as important to keep your youthful questions alive. [19]

this is one of the places where actual expertise differs most from the popular picture of it. in the popular picture, experts are certain. but actually the more puzzled you are, the better, so long as (a) the things you're puzzled about matter, and (b) no one else understands them either.

think about what's happening at the moment just before a new idea is discovered. often someone with sufficient expertise is puzzled about something. which means that originality consists partly of puzzlement — of confusion! you have to be comfortable enough with the world being full of puzzles that you're willing to see them, but not so comfortable that you don't want to solve them. [20]

though conservatism is usually associated with the old, it's the young who tend to make this mistake. inexperience makes them fear risk, but it's when you're young that you can afford the most.

even a project that fails can be valuable. in the process of working on it, you'll have crossed territory few others have seen, and encountered questions few others have asked. and there's probably no better source of questions than the ones you encounter in trying to do something slightly too hard.

use the advantages of youth when you have them, and the advantages of age once you have those. the advantages of youth are energy, time, optimism, and freedom. the advantages of age are knowledge, efficiency, money, and power. with effort you can acquire some of the latter when young and keep some of the former when old.

the old also have the advantage of knowing which advantages they have. the young often have them without realizing it. the biggest is probably time. the young have no idea how rich they are in time. the best way to turn this time to advantage is to use it in slightly frivolous ways: to learn about something you don't need to know about, just out of curiosity, or to try building something just because it would be cool, or to become freakishly good at something.

that "slightly" is an important qualification. spend time lavishly when you're young, but don't simply waste it. there's a big difference between doing something you worry might be a waste of time and doing something you know for sure will be. the former is at least a bet, and possibly a better one than you think. [23]

the most subtle advantage of youth, or more precisely of inexperience, is that you're seeing everything with fresh eyes. when your brain embraces an idea for the first time, sometimes the two don't fit together perfectly. usually the problem is with your brain, but occasionally it's with the idea. a piece of it sticks out awkwardly and jabs you when you think about it. people who are used to the idea have learned to ignore it, but you have the opportunity not to. [24]

so when you're learning about something for the first time, pay attention to things that seem wrong or missing. you'll be tempted to ignore them, since there's a 99% chance the problem is with you. and you may have to set aside your misgivings temporarily to keep progressing. but don't forget about them. when you've gotten further into the subject, come back and check if they're still there. if they're still viable in the light of your present knowledge, they probably represent an undiscovered idea.

one of the most valuable kinds of knowledge you get from experience is to know what you *don't* have to worry about. the young know all the things that could matter, but not their relative importance. so they worry equally about everything, when they should worry much more about a few things and hardly at all about the rest.

but what you don't know is only half the problem with inexperience. the other half is what you do know that ain't so. you arrive at adulthood with your head full of nonsense — bad habits you've acquired and false things you've been taught — and you won't be able to do great work till you clear away at least the nonsense in the way of whatever type of work you want to do.

much of the nonsense left in your head is left there by schools. we're so used to schools that we unconsciously treat going to school as identical with learning, but in fact schools have all sorts of strange qualities that warp our ideas about learning and thinking.

for example, schools induce passivity. since you were a small child, there was an authority at the front of the class telling all of you what you had to learn and then measuring whether you did. but neither classes nor tests are intrinsic to learning; they're just artifacts of the way schools are usually designed.

some of the features of things you admire are flaws they succeeded despite. indeed, the features that are easiest to imitate are the most likely to be the flaws.

this is particularly true for behavior. some talented people are jerks, and this sometimes makes it seem to the inexperienced that being a jerk is part of being talented. it isn't; being talented is merely how they get away with it.

one of the most powerful kinds of copying is to copy something from one field into another. history is so full of chance discoveries of this type that it's probably worth giving chance a hand by deliberately learning about other kinds of work. you can take ideas from quite distant fields if you let them be metaphors.

negative examples can be as inspiring as positive ones. in fact you can sometimes learn more from things done badly than from things done well; sometimes it only becomes clear what's needed when it's missing.

if a lot of the best people in your field are collected in one place, it's usually a good idea to visit for a while. it will increase your ambition, and also, by showing you that these people are human, increase your self-confidence. [26]

if you're earnest you'll probably get a warmer welcome than you might expect. most people who are very good at something are happy to talk about it with anyone who's genuinely interested. if they're really good at their work, then they probably have a hobbyist's interest in it, and hobbyists always want to talk about their hobbies.

it may take some effort to find the people who are really good, though. doing great work has such prestige that in some places, particularly universities, there's a polite fiction that everyone is engaged in it. and that is far from true. people within universities can't say so openly, but the quality of the work being done in different departments varies immensely. some departments have people doing great work; others have in the past; others never have.

seek out the best colleagues. there are a lot of projects that can't be done alone, and even if you're working on one that can be, it's good to have other people to encourage you and to bounce ideas off.

colleagues don't just affect your work, though; they also affect you. so work with people you want to become like, because you will.

quality is more important than quantity in colleagues. it's better to have one or two great ones than a building full of pretty good ones. in fact it's not merely better, but necessary, judging from history: the degree to which great work happens in clusters suggests that one's colleagues often make the difference between doing great work and not.

how do you know when you have sufficiently good colleagues? in my experience, when you do, you know. which means if you're unsure, you probably don't. but it may be possible to give a more concrete answer than that. here's an attempt: sufficiently good colleagues offer *surprising* insights. they can see and do things that you can't. so if you have a handful of colleagues good enough to keep you on your toes in this sense, you're probably over the threshold.

most of us can benefit from collaborating with colleagues, but some projects require people on a larger scale, and starting one of those is not for everyone. if you want to run a project like that, you'll have to become a manager, and managing well takes aptitude and interest like any other kind of work. if you don't have them, there is no middle path: you must either force yourself to learn management as a second language, or avoid such projects. [27]

ing and walking are particularly good forms of exercise because they're good for thinking. [29]

people who do great work are not necessarily happier than everyone else, but they're happier than they'd be if they didn't. in fact, if you're smart and ambitious, it's dangerous *not* to be productive. people who are smart and ambitious but don't achieve much tend to become bitter.

it's ok to want to impress other people, but choose the right people. the opinion of people you respect is signal. fame, which is the opinion of a much larger group you might or might not respect, just adds noise.

the prestige of a type of work is at best a trailing indicator and sometimes completely mistaken. if you do anything well enough, you'll make it prestigious. so the question to ask about a type of work is not how much prestige it has, but how well it could be done.

competition can be an effective motivator, but don't let it choose the problem for you; don't let yourself get drawn into chasing something just because others are. in fact, don't let competitors make you do anything much more specific than work harder.

curiosity is the best guide. your curiosity never lies, and it knows more than you do about what's worth paying attention to.

notice how often that word has come up. if you asked an oracle the secret to doing great work and the oracle replied with a single word, my bet would be on "curiosity."

that doesn't translate directly to advice. it's not enough just to be curious, and you can't command curiosity anyway. but you can nurture it and let it drive you.

curiosity is the key to all four steps in doing great work: it will choose the field for you, get you to the frontier, cause you to notice the gaps in it, and drive you to explore them. the whole process is a kind of dance with curiosity.

believe it or not, i tried to make this essay as short as i could. but its length at least means it acts as a filter. if you made it this far, you must be interested in doing great work. and if so you're already further along than you might realize, because the set of people willing to want to is small.

the factors in doing great work are factors in the literal, mathematical sense, and they are: ability, interest, effort, and luck. luck by definition you can't do anything about, so we can ignore that. and we can assume effort, if you do in fact want to do great work. so the problem boils down to ability and interest. can you find a kind of work where your ability and interest will combine to yield an explosion of new ideas?

here there are grounds for optimism. there are so many different ways to do great work, and even more that are still undiscovered. out of all those different types of work, the one you're most suited for is probably a pretty close match. probably a comically close match. it's just a question of finding it, and how far into it your ability and interest can take you. and you can only answer that by trying.

many more people could try to do great work than do. what holds them back is a combination of modesty and fear. it seems presumptuous to try to be newton or shakespeare. it also seems hard; surely if you tried something like that, you'd fail. presumably the calculation is rarely explicit. few people consciously decide not to try to do great work. but that's what's going on subconsciously; they shy away from the question.

so i'm going to pull a sneaky trick on you. do you want to do great work, or not? now you have to decide consciously. sorry about that. i wouldn't have done it to a general audience. but we already know

10 if you set your life up right, it will deliver the focus-relax cycle automatically. the perfect setup is an office you work in and that you walk to and from.

11 there may be some very unworldly people who do great work without consciously trying to. if you want to expand this rule to cover that case, it becomes: don't try to be anything except the best.

12 this gets more complicated in work like acting, where the goal is to adopt a fake persona. but even here it's possible to be affected. perhaps the rule in such fields should be to avoid *unintentional* affectation.

13 it's safe to have beliefs that you treat as unquestionable if and only if they're also unfalsifiable. for example, it's safe to have the principle that everyone should be treated equally under the law, because a sentence with a "should" in it isn't really a statement about the world and is therefore hard to disprove. and if there's no evidence that could disprove one of your principles, there can't be any facts you'd need to ignore in order to preserve it.

14 affectation is easier to cure than intellectual dishonesty. affectation is often a shortcoming of the young that burns off in time, while intellectual dishonesty is more of a character flaw.

15 obviously you don't have to be working at the exact moment you have the idea, but you'll probably have been working fairly recently.

16 some say psychoactive drugs have a similar effect. i'm skeptical, but also almost totally ignorant of their effects.

17 for example you might give the nth most important topic (m-1)/m^n of your attention, for some m > 1. you couldn't allocate your attention so precisely, of course, but this at least gives an idea of a reasonable distribution.

18 the principles defining a religion have to be mistaken. otherwise anyone might adopt them, and there would be nothing to distinguish the adherents of the religion from everyone else.

19 it might be a good exercise to try writing down a list of questions you wondered about in your youth. you might find you're now in a position to do something about some of them.

20 the connection between originality and uncertainty causes a strange phenomenon: because the conventional-minded are more certain than the independent-minded, this tends to give them the upper hand in disputes, even though they're generally stupider.

> the best lack all conviction, while the worst
> are full of passionate intensity.

21 derived from linus pauling's "if you want to have good ideas, you must have many ideas."

22 attacking a project as a "toy" is similar to attacking a statement as "inappropriate." it means that no more substantial criticism can be made to stick.

23 one way to tell whether you're wasting time is to ask if you're producing or consuming. writing computer games is less likely to be a waste of time than playing them, and playing games where you create something is less likely to be a waste of time than playing games where you don't.

24 another related advantage is that if you haven't said anything publicly yet, you won't be biased toward evidence that supports your earlier conclusions. with sufficient integrity you could achieve eternal youth in this respect, but few manage to. for most people, having previously published opinions has an effect similar to ideology, just in quantity 1.

25 in the early 1630s daniel mytens made a painting of henrietta maria handing a laurel wreath to charles i. van dyck then painted his own version to show how much better he was.

26 i'm being deliberately vague about what a place is. as of this writing, being in the same physical place has advantages that are hard to duplicate, but that could change.

27 this is false when the work the other people have to do is very constrained, as with seti@home or bitcoin. it may be possible to expand the area in which it's false by defining similarly restricted protocols with more freedom of action in the nodes.